



Ecological specialization matters: long-term trends in butterfly species richness and assemblage composition depend on multiple functional traits

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ABSTRACT

Aim To quantify spatio-temporal changes in species richness and assemblage composition of Danish butterflies over more than 100 years and to assess whether different functional groups of butterflies show different temporal trajectories.

Location Denmark, Europe.

Methods We applied species accumulation curves to estimate relative species richness changes from unequally sampled occurrence records, collected over more than a century (1900–2012), at local (i.e. 10 km × 10 km) and regional (country-wide) spatial scales. Furthermore, we calculated similarity in species composition between 10 km × 10 km grid cells to evaluate changes in assemblage composition (i.e. biotic homogenization) of butterflies through space and time. Trends were evaluated for all butterflies as well as for three functional groups, using hierarchical clustering and species-specific values of eight ecological traits.

Results We document severe declines in butterfly diversity over the last century, with substantial regional-scale extinctions leading to a net loss of 10% of all Danish butterfly species. Segregating species into functional groups showed that the highest rate of regional extinction occurred among sedentary habitat specialists overwintering in the egg stage, while the most severe local-scale declines occurred among sedentary host plant specialists overwintering in the larval stage. In contrast, mobile generalist species with mature overwintering stages showed stable richness trends throughout the century. However, a pervasive spatio-temporal biotic homogenization of butterfly assemblages was revealed for all functional groups.

Main conclusions Our results suggest that observed changes in species richness were driven by a gradual replacement of ecological specialists by broadly adapted ecological generalists. In addition to drawing attention to a severe long-term impoverishment of the Danish butterfly fauna, our results suggest that trajectories of butterfly species richness and assemblage composition over the last century are related to multiple functional traits, highlighting the need to consider different aspects of ecological specialization when assessing extinction risk.

Keywords

Assemblage homogenization, butterflies, extinction, functional traits, global change, species richness decline.

INTRODUCTION

Ongoing global change is driving biodiversity loss and profoundly altering the properties of ecosystems (Parmesan, 2006; Pereira *et al.*, 2010; Potts *et al.*, 2010). The result is likely to be a more homogenized biosphere, where ecological specialists are gradually replaced by fewer widespread and broadly adapted ecological generalists (McKinney & Lockwood, 1999; Olden *et al.*, 2004; Börschig *et al.*, 2013). Consequently, critical ecosystem processes (e.g. pollination) may be constrained, reducing ecosystem resilience to further environmental change (Schröter *et al.*, 2005; Bartomeus *et al.*, 2013). In addition to losing alpha- and beta-diversity, the composition of functional diversity in communities may shift through time (McKinney & Lockwood, 1999; Biesmeijer *et al.*, 2006; Börschig *et al.*, 2013). This is expected because species' responses to changes in their environment often depend on life history characteristics (Henle *et al.*, 2004; Öckinger *et al.*, 2010; Diamond *et al.*, 2011). Thus, identifying the key characteristics which determine organisms' sensitivity to environmental change is critical to ecological forecasting and conservation planning. However, very few datasets extend back far enough to assess such differences in the temporal dynamics among functional groups of species.

Species with specialized habitat requirements, narrow environmental tolerances and poor dispersal ability are likely to be particularly susceptible to global change (Henle *et al.*, 2004; Pöyry *et al.*, 2009; Öckinger *et al.*, 2010; Diamond *et al.*, 2011). Also, historical land use changes may define diverging population trajectories for species with different ecological niches (Krauss *et al.*, 2010). Thus, species belonging to separate functional groups might show contrasting responses to global change, and any potential local or regional extinction debt associated with these changes may be 'paid back' at different rates (Krauss *et al.*, 2010). Conversely, species with traits that promote establishment under changed conditions may show positive population responses to global change (McKinney & Lockwood, 1999). Research on the relationship between diversity and functional aspects has so far been dominated by studies that focus on the dichotomy between resource specialists and generalists (Menéndez *et al.*, 2006; Stefanescu *et al.*, 2011; Börschig *et al.*, 2013; but see Dapporto & Dennis, 2013). Often, butterflies are classified as either generalists (broad) or specialists (narrow) in their resource use, typically distinguished by larval dietary breadth (Dennis, 2012). Yet, the environmental responses of species are possibly related to a complex array of different traits, for example host plant or habitat specialization, reproductive rate or mobility (Diamond *et al.*, 2011; Börschig *et al.*, 2013). Furthermore, a species may be a specialist on one resource, for example larval host plants, but a generalist for another, for example habitat type (Loxdale *et al.*, 2011; Dennis, 2012; Dapporto & Dennis, 2013). So far, it remains largely unclear whether different ecological specializations may lead to different long-term responses to global changes.

Documenting and understanding the historical spatio-temporal dynamics of biotic communities is often challenged by the lack of suitable data. In the absence of long-term standardized surveys (e.g. monitoring datasets), public and private specimen collections are a valuable source of information about past and present species occurrences (Miller-Rushing *et al.*, 2006). To deal with bias and unequal sampling associated with such haphazardly collected datasets, new methods have been developed to estimate species richness change (alpha-diversity) (Biesmeijer *et al.*, 2006; Keil *et al.*, 2011; Bartomeus *et al.*, 2013) and to evaluate spatio-temporal homogenization (decrease in beta-diversity) of communities (Carvalho *et al.*, 2013). These methods include estimations based on species accumulation curves to standardize comparisons with the least sampled period. This provides robust estimates of changes in species richness and assemblage composition across long ecological time-scales (e.g. decades or centuries).

Ecological and distributional knowledge of butterflies (Lepidoptera: Papilionoidea) is well represented due to their charismatic nature and attractiveness for professional and amateur naturalists (Wilson *et al.*, 2004; Thomas, 2005; Devictor *et al.*, 2012). Butterflies exhibit a diverse range of habitat requirements, dietary preferences and dispersal abilities (Thomas, 2005). This, combined with their high sensitivity to climate change, habitat degradation and fragmentation make them an important taxonomic group for early detection of the consequences of global change for biodiversity (Thomas, 2005). Here, we compiled and analyzed a new, unstandardized historical dataset with > 400,000 Danish butterfly occurrences covering > 100 years and applied recently developed analytical methods to detect long-term changes in species richness and assemblage composition at two spatial scales (local, i.e. 10 km × 10 km, and regional, i.e. country-wide). We further evaluated differences in species richness and assemblage composition change among functional groups of butterflies based on eight functional traits (migratory behaviour, body size, larval dietary breadth, host plant availability, habitat specialization, habitat type, overwintering strategy and voltinism). We ask whether different functional groups of butterflies show different temporal trajectories, hypothesizing that functional groups with narrow ecological adaptations (e.g. high resource specialization, low mobility and low reproductive rate) show stronger evidence of richness declines and biotic homogenization than groups with broader and more flexible ecological adaptations.

METHODS

Butterfly occurrence data

A comprehensive dataset with > 427,000 geo-referenced observations of Danish butterfly occurrences was collated from two entomological societies, two national atlas surveys, two major citizen science databases and one private specimen collection (Table S1). Fifteen vagrant species, which have

been recorded in Denmark, but have never had breeding populations in the country, were not included in our analyses (Table S2). Migrant species, which breed in Denmark regularly, were kept in the analysis. The final validated dataset spanned 113 years (1900–2012) and represented 79 butterfly species and 358,663 unique observations. For the richness change analyses, records were binned into five 20-year time slices and one final 13-year time slice (1900–19, 1920–39, 1940–59, 1960–79, 1980–99 and 2000–12). Changes in richness were compared between each of the five 20-year time slices and the final time slice (2000–12).

Analysis of species richness change

Estimates of species richness are highly dependent on sampling effort (e.g. number of records per grid cell and time period). Species accumulation curves (providing estimates of expected richness with increasing number of records) can then be used to estimate species richness for a given sampling effort (Gotelli & Colwell, 2001). This allows comparison of species richness estimates among regions or between time periods with unequal sampling effort. We followed the methods described in Carvalheiro *et al.* (2013) to deal with such particularities of historical datasets when estimating richness changes. This allowed us to deal with over-representation of rare species and with under-representation of singletons (i.e. species represented by only a single record in a given cell) which could result from collection efforts being concentrated to represent morphological diversity. As in Carvalheiro *et al.* (2013), we only estimated richness change for cells that were considered well sampled, that is that matched the following selection criteria: (1) cells had more than 15 records per time period, (2) the ratio of records/number of species was higher than 1.5 in each of the two compared time periods and (3) if number of records was less than fivefold the total number of species in the country, we only considered cells with less than a 10-fold difference in numbers of records between periods. Applying less strict selection criteria would have led to the selection of more grid cells, but also an increased risk of making type II errors (Carvalheiro *et al.*, 2013). An overview of the total number of records and grid cells used per time slice is provided in Table S3 and S4, respectively.

After applying these selection criteria, we calculated an overall weighted value of relative richness change between each of the five 20-year time slices and the final time slice (2000–12). We estimated change based on the extrapolation of the species richness of the least sampled period up to the larger sample size, which when dealing with randomly sampled data provides reliable results up to about triple the original number of the records (Colwell *et al.*, 2012). Where the most sampled period had a sampling effort larger than threefold the number of records in the least sampled period, we used both extrapolation and rarefaction in our comparisons: the assessment of richness change was then made at the point on the accumulation curve where sampling effort was

three times the number of records of the least sampled period, with the small sample then extrapolated up to this point, and the larger sample rarefied down to it (Carvalheiro *et al.*, 2013). Subsequently, we accounted for the effect of over-/underrepresentation of singletons using bootstrapping to calculate the variance of richness change estimates of each cell. We then weighted each cell by the inverse of its variance to obtain an overall weighted value of richness change and to assess if such a value was significantly different from zero (Carvalheiro *et al.*, 2013). We used 95% confidence intervals generated around the overall weighted value of richness change for each time slice to assess whether the mean value of richness in each time slice was different from the baseline (i.e. confidence intervals not overlapping with 100), and whether species richness between subsequent time periods was different (i.e. confidence intervals between neighbouring periods not overlapping with each other) (Table S5). All calculations described here and below were performed in R version 2.15.1 (R Development Core Team, 2013). For the richness change analyses, we used functions from the packages 'BOOT', 'METAPHOR' and 'NLME'. It should be further noted that the species turnovers reported here are conservative and could potentially mask even larger changes within the butterfly assemblages because simultaneous losses and gains of species at a given spatial scale might cancel each other out. A more detailed description of the methods is provided in Carvalheiro *et al.* (2013).

Estimates of species richness change can be strongly influenced by the spatial resolution at which they are performed, affecting the magnitude and sometimes even the direction of the estimated change (Keil *et al.*, 2011). We therefore performed all analyses of species richness change at both a local (i.e. 10 km × 10 km grid cells) and regional resolution (all of Denmark).

Ecological traits and functional grouping

Eight species trait categories, which have been suggested to be important for butterfly responses to global change (Henle *et al.*, 2004; Pöyry *et al.*, 2009; Öckinger *et al.*, 2010; Diamond *et al.*, 2011), were included in the analysis. These traits were extracted from the literature on Danish butterflies (Stoltze, 1996; García-Barros, 2000; Frederiksen *et al.*, 2006) and represented overwintering stage, migratory behaviour, degree of habitat openness, habitat specialization, larval host plant distribution, voltinity (number of generations per year), body size and host plant specialization (larval dietary breadth). We defined trait categories using local knowledge, that is 'realized' niches. Even if a species can be considered a generalist on the European scale, but a specialist in Denmark (due to missing alternatives or different ecological conditions), it is the loss of these particular resources (i.e. habitat types or food plants) which makes a species decline in Denmark, regardless of the potential at larger scales. A detailed overview of each trait category and its respective states is provided in Table 1. Species-level data are provided in Table S6. Species were

Table 1 Eight species traits of Danish butterflies. Trait categories were subjected to a Hill & Smith principal component analysis and the correlation ratio for each trait category is shown for axes 1 and 2. Traits are ordered according to their importance, with numbers in bold showing the ratios that are higher than the average correlation ratio for that axis (mean correlation ratios: axis 1 = 0.40; axis 2 = 0.31). Traits 1–4, 6 and 8 are taken from Stoltze (1996), 5 from Frederiksen *et al.* (2006) and 7 from García-Barros (2000). Species-specific trait values can be found in Table S6.

Trait category	Description of trait states	Axis 1	Axis 2
Overwintering stage	'Egg', 'larvae', 'pupae', 'adult'	0.58	0.43
Habitat openness	Average habitat vegetation openness score for each butterfly species based on an openness scores (1–3) for nine habitat types: rocky outcrop, coastal/dunes, grassland, heath, mire and field margin (score 1), urban (score 2) and forest edges and forest meadows (score 3)	0.31	0.52
Migratory behaviour	'Migratory', 'non-migratory'	0.67	0.13
Habitat specialization	Species classified as either 'specialists' (using only one habitat type), 'semi-specialists' (using two different habitat types), or 'generalists' (using more than three or more different habitat types), based on the nine habitat types mentioned above	0.33	0.47
Host plant strribution	The distribution of each butterfly species' host plant species in Denmark ('widespread' or 'local'). For oligo- or polyphagous butterflies, only the distribution of the most widespread host plant was considered	0.40	0.35
Body size	Average forewing length of both males and females (measured in mm)	0.27	0.35
Host plant specialization	Species classified as either 'monophagous' (only using one host plant species), 'oligophagous' (using several host plant species in the same genus) or 'polyphagous' (using host plants in multiple genera)	0.26	0.30
Voltinity	Number of generations per year	0.38	0.03

separated into functional groups based on the trait states of each trait category (Schweiger *et al.*, 2007). We first applied ordination methods to produce species-specific values on orthogonal trait axes and then used hierarchical clustering to group all butterflies into functional groups based on these species-specific values. For the ordination, we applied a Hill & Smith principal component analysis (Hill & Smith, 1976) which is appropriate for analyses that contain both categorical and continuous species trait states. This was done using the 'dudi.hillsmith' function of the R package 'ADE4'. Based on the Cattell's scree test (Cattell, 1966), we retained the first two ordination axes in the analysis and applied a hierarchical cluster analysis with the unweighted pair-group method using arithmetic averages (UPGMA) to produce a dendrogram of the trait similarities among species. This was done with the 'daisy' function of the R package 'CALIBRATE' and the 'AGNES' function of the 'CLUSTER' package. The optimal number of clusters in the dendrogram was then determined with the Calinski–Harabasz index (Caliński & Harabasz, 1974) using the 'calinhara' function of the R package 'FPC'. Four functional groups (clusters) were identified (Fig. S1). Due to its small size, the fourth functional group had to be excluded from further analyses and the analysis of species richness change was subsequently run only for the three largest functional groups, following the same steps as described above.

Analysis of assemblage composition through space and time

To understand how the composition of butterfly communities has changed through space and time, we calculated

similarity in species composition between each focal grid cell and all other 10 km × 10 km cells. We used similarity due to species replacement (calculated as 1 minus beta-sim, Lennon *et al.*, 2001), a measure that is independent of, but complementary to, our measurements of alpha-diversity (see Baselga, 2010). To correct for the unequal sampling effort, we used an individual-based probabilistic approach using the formulas described in Chao *et al.* (2005). As species identity is crucial for similarity analyses, we applied selection criteria that were stricter than those of the richness analyses. Thus, only those cells were included that had (1) records per grid cell that were equal to or higher than the maximum number of species possible to find in each cell within a given time period and (2) a ratio of records/number of species higher than 1.5. We analyzed for each time period (and each functional group) how similarity (after logit transformation to standardize residuals) varied with distance between grid cells, that is quantifying the distance decay of similarity (Nekola & White, 1999). We fitted generalized linear mixed models (with focal grid cell as a random factor) using restricted maximum likelihood in the R package 'NLME'. The statistical analyses followed Carvalheiro *et al.* (2013) and statistical details of the fitted models are provided in Table S7 for the different time periods and different functional groups.

RESULTS

Overall richness change at local and regional scales

Overall butterfly species richness was stable at both local and regional scales during the first 80 years of the study period

(Fig. 1a). However, this was followed by a severe decline in richness at both scales between 1960–79 and 1980–99. A less accentuated decline was subsequently seen at both scales between 1980–99 and 2000–12. This resulted in a net loss of 10% of all Danish butterfly species at the regional scale and a 20% net loss of species richness at the local scale within a time frame of five decades. All extinct species are listed in Table S6.

Functional grouping of butterflies

The multivariate ordination of the eight species traits revealed several relationships among traits (Fig. S1). The first two factorial axes of the ordination explained 35.5% of the total variance in species traits and were retained in further analyses. Overwintering stage contributed most to species separation, that is had higher than average correlation values for both factorial axes (Table 1). Furthermore, habitat openness, migratory behaviour, habitat specialization, host plant distribution and body size had higher than average correlation ratios with one of the two ordination axes. Voltinity and host plant specialization showed low correlation values with both axes, thus contributing little to species separation (Table 1).

The hierarchical clustering resulted in the identification of four clearly separated functional groups, consisting of

23, 30, 22 and four species, respectively (Table 2, Fig. S1). The fourth group was excluded from further analyses due to its small size. Several trait states showed higher than average correlation ratios with these clusters, indicating strong contributions to species separation into functional groups (shown in bold in Table 2). Species in clusters 1 and 2 were characterized by high ecological specialization. Species in cluster 1 were primarily characterized by specialization to few and locally distributed host plants, but did not show a strong habitat specialization as they were associated with several different types of open habitats (e.g. grassland, heath). In contrast, species in cluster 2 were characterized by strong habitat specialization, especially to forest habitat, but were less restricted in their host plant use. Both groups were characterized by low mobility (i.e. non-migratory) and overwintering in immature stages (egg, larvae). Unlike these two specialist groups, species in cluster 3 had mature overwintering stages (adult, pupa), were highly mobile (i.e. migratory) and showed no particular host plant- or habitat specialization. The three clusters are subsequently referred to as ‘sedentary host plant specialists’ (cluster 1), ‘sedentary habitat specialists’ (cluster 2) and ‘mobile generalists’ (cluster 3). Although simplistic, these names are meant to highlight some of the key ecological traits of each functional group.

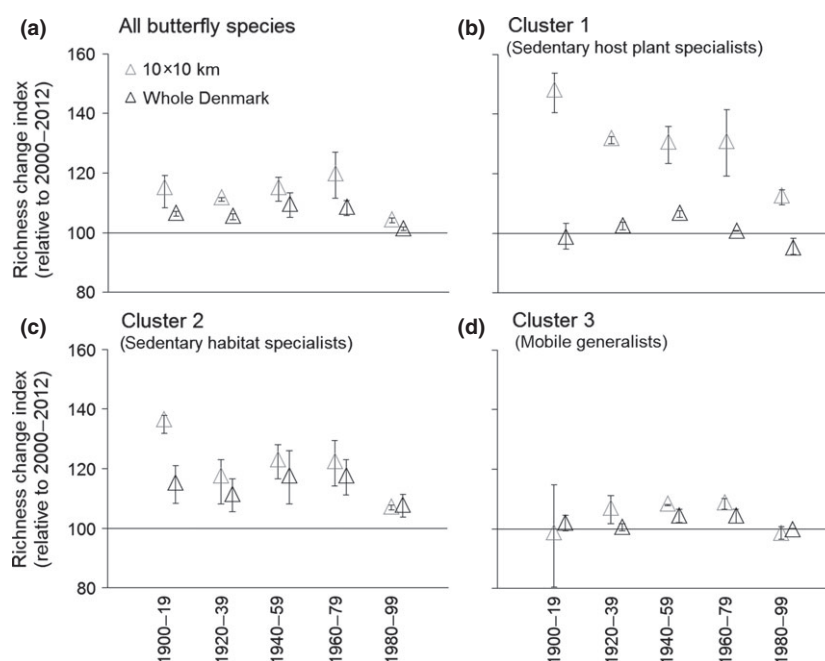


Figure 1 Species richness changes (estimated weighted mean $\pm$ 95% confidence intervals) for (a) all Danish butterfly species, (b) cluster 1 (sedentary host plant specialists overwintering in the larval stage) (c) cluster 2 (sedentary habitat specialists overwintering in the egg stage) and (d) cluster 3 (mobile generalists with mature overwintering stages). The three functional groups (b–c) correspond to clusters 1–3 in Table 2, which were obtained using hierarchical clustering and species-specific values of eight ecological traits (Table 1). Observed distribution records were initially binned into six time slices (1900–19, 1920–39, 1940–59, 1960–79, 1980–99 and 2000–12) and richness changes were then estimated for each time slice relative to the most recent time slice (i.e. 2000–12). The most recent time slice serves as a baseline for comparison among time slices with its value being set at 100. Richness changes are shown for two different spatial resolutions (10 km $\times$ 10 km and all of Denmark).

Table 2 Functional grouping of butterfly species. Three different clusters (cluster 1–3) are distinguished by means of hierarchical clustering (Fig. S1) on the basis of factorial loads resulting from an ordination analysis of eight different species trait categories (Table 1). The number of species per functional group is indicated in brackets. Traits with higher-than-average correlation ratios (and thus contributing strongly to the species separation) are marked in bold.

Trait category	Cluster 1 (n = 23)	Cluster 2 (n = 30)	Cluster 3 (n = 22)
Overwintering stage	Larvae	Egg	Adult/pupa
Migratory behaviour	No	No	Yes
Habitat openness	Open	Closed	Intermediate
Habitat specialization	Intermediate	Specialist	Generalist
Host plant distribution	Local	Widespread	Widespread
Voltinity	Few	Few	Many
Body size	Small	Intermediate	Large
Host plant specialization	Monophagous	Oligophagous	Polyphagous

Richness change of functional groups

For all three functional groups, we detected substantial richness losses during the past century (Fig. 1). However, trends differed markedly between each group (Fig. 1b–d, Table S5). Sedentary host plant specialists exhibited the strongest declines at the local spatial resolution (Fig. 1b). A first richness loss period occurred between 1900–19 and 1920–39, and a second between 1960–79 and 2000–12. Regional extinctions were low and occurred mostly between the 1940s and the 1980s, with a slight recovery in species richness in recent years. Sedentary habitat specialists (Fig. 1c) showed strong richness declines at local resolution during the past 100 years. After 1960–79, these were accompanied by severe regional extinctions, indicated by a drop in regional-scale richness (Fig. 1c). This resulted in a total net loss of 27% of species in this group. For mobile generalists (Fig. 1d), species richness remained stable at local resolution during most of the century, and regional extinctions remained low.

Distance decay of assemblage similarity through time

When comparing the distance decay curves for similarity of overall butterfly richness across the six time periods, an increase in both intercept values (i.e. adjacent species assemblages becoming more similar, indicating local homogenization) as well as slope values (i.e. distant species assemblages becoming more similar, indicating regional-scale homogenization) became apparent (Fig. 2a). Between 1940–59 and 1980–99, significant differences in intercepts and slopes of

distance decay curves for overall butterfly richness between subsequent time periods (Fig. 2a, Table S7) revealed a strong spatial homogenization of Danish butterfly assemblages at both local and regional scales. No further homogenization was detected between 1980–99 and 2000–12.

Marked differences in rates of homogenization were found for the three functional groups (Fig. 2b–d, Table S7). Sedentary host plant specialists showed an increase in local assemblage similarity (increasing intercept values) between 1940–59 and 1960–79, followed by a wave of strong large-scale homogenization (increasing slope values) between 1960–79 and 2000–12 (Fig. 2b). Sedentary habitat specialists showed strong local assemblage homogenization (increasing intercept values) between 1940–59 and 1989–99 (Fig. 2c), but the degree of large-scale homogenization (increasing slope values) was less accentuated than for the other functional groups. For mobile generalists, the degree of assemblage similarity was already high from the beginning of the century (i.e. high intercept values and low distance decay slope values). Nevertheless, significant increases in slopes and intercepts were apparent between 1940–59 and 1980–99, indicating further assemblage homogenization for this functional group. These signals of increasing biotic homogenization even in generalist assemblages are noteworthy because regional richness changes in this functional group were small (Fig. 1d versus Fig. 2d).

DISCUSSION

Overall trends in species richness

Evaluating how rates of biodiversity loss are changing through space and time is essential for ecological forecasting and conservation planning. By compiling and analysing a large dataset of Danish butterfly occurrences covering > 100 years, we revealed substantial declines in butterfly diversity over the last century at local and regional scales, ultimately leading to a net loss of 10% of all Danish butterfly species.

Differences in richness change of functional groups

Our analysis showed that butterfly species could be segregated into three clearly separated functional groups, which showed marked differences in the strength, timing and spatial scale of richness change. The highest rate of regional extinction occurred among sedentary habitat specialists overwintering in the egg stage (cluster 2), while the most severe local-scale losses occurred among sedentary host plant specialists overwintering in the larval stage (cluster 1). In contrast, mobile species with broad ecological adaptations and mature overwintering stages (cluster 3) were stable throughout the century.

In addition to drawing attention to a severe long-term impoverishment of the Danish butterfly fauna, our results suggest that trajectories of butterfly species diversity and

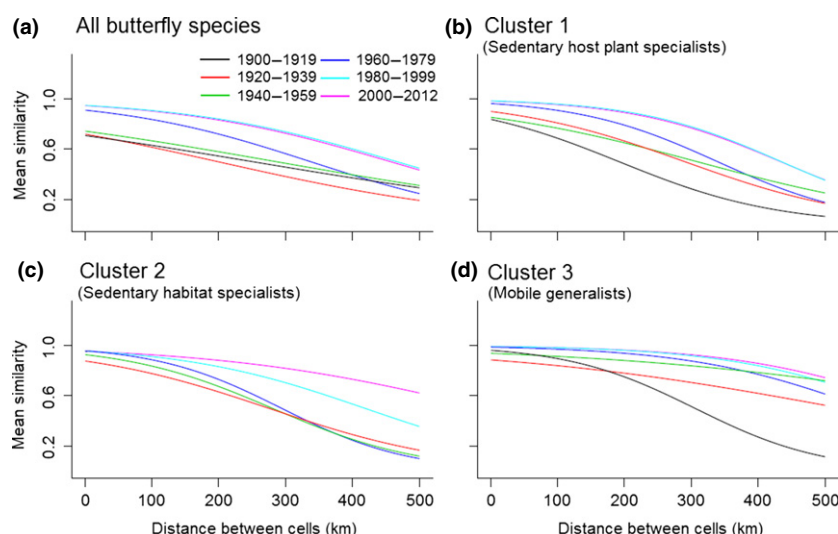


Figure 2 Changes in similarity of species composition of butterfly assemblages through space and time for (a) all Danish butterfly species, (b) cluster 1 (sedentary host plant specialists overwintering in the larval stage), (c) cluster 2 (sedentary habitat specialists overwintering in the egg stage) and (d) cluster 3 (mobile generalists with mature overwintering stages). The three functional groups (b–c) correspond to clusters 1–3 in Table 2, which were obtained using hierarchical clustering and species-specific values of eight ecological traits (Table 1). Lines represent the estimated value of similarity between 10 km $\times$ 10 km grid cells after removing the effect of sampling effort (see Methods). Statistical details of the fitted models are provided in Table S7. A temporal pattern of increasing intercepts (i.e. adjacent species assemblages becoming more similar) and declining slopes (i.e. distant species assemblages becoming more similar) indicates increasing assemblage similarity at both local and regional scales through time, with the most homogenous butterfly assemblages occurring in the most recent time periods.

assemblage composition at a centennial time-scale are related to a complex array of different traits. Our study therefore highlights the importance of addressing different aspects of ecological specialization in future studies. By segregating our species into functional groups, it became apparent that extensive species richness loss occurred for highly specialized functional groups already from the early 1900s, even though this was not immediately apparent from the species richness analysis of the full dataset (Fig. 1b and Fig. 1c versus Fig. 1a). This result is particularly noteworthy from a conservation perspective, because it indicates that traditional species richness studies, which do not segregate species into functional groups, may overlook important differences in the strength and timing of richness change of species with different ecological adaptations.

Severe functional homogenization of Danish butterfly assemblages

Species richness declines were accompanied by a pervasive spatio-temporal biotic homogenization of butterfly assemblages, suggesting that the functional composition of Danish butterfly assemblages has changed considerably, not only through the local and regional loss of ecological specialists (i.e. clusters 1 and 2), but also through the simultaneous spatio-temporal homogenization of generalist assemblages (i.e. cluster 3). In particular, the signals of increasing biotic homogenization in generalist assemblages are noteworthy because regional richness changes in this functional group

were small (Fig. 1d versus Fig. 2d). Overall, these results highlight that many species richness studies, which report stable trends, may be missing this important aspect of biotic homogenization in generalists. This could be of major concern for conservation management and planning.

Trend for sedentary host plant specialists overwintering in the larval stage (cluster 1)

Our results indicate that it is the distribution of host plants rather than the degree of host plant specialization alone that is the important factor for butterfly species richness changes (Table 2). In the late 1800s, semi-natural grassland covered 25% of Denmark, but agricultural intensification has since reduced the areal extent to < 10% (Levin & Normander, 2008). Similar trends have been documented in other European countries, where they have been suggested as drivers of severe butterfly declines (Kuussaari *et al.*, 2007; Van Dyck *et al.*, 2009; Sang *et al.*, 2010; European Environmental Agency, 2013). The low mobility of sedentary host plant specialists, combined with their reliance on locally distributed host plant resources, is likely to have rendered this group particularly vulnerable to the loss and fragmentation of semi-natural habitats. Richness declines for this functional group demonstrate that strong, local-scale deterioration of butterfly communities was apparent already in the beginning of the 1900s (Fig. 1b, grey symbols). This was accompanied by a trend towards severe homogenization of butterfly assemblages at both local and regional scales (Fig. 2b).

In contrast to sedentary habitat specialists, regional extinctions among host plant specialists were rare (eight species versus two species, respectively). This may be explained by the historically higher abundance of semi-natural habitats compared to forest (the predominant habitat type of sedentary habitat specialists species) in Denmark (Levin & Normander, 2008). Yet, considering the severity of local-scale losses found for this group, assemblages are likely to have accumulated an unpaid extinction debt. Population declines are known to lag considerably behind the socioeconomic pressures that drive them (Dullinger *et al.*, 2013), with delays in insect metapopulation extinctions sometimes extending well beyond 100 years (Bulman *et al.*, 2007). These lag times may be particularly long if available habitat networks are fragmented, but still maintain populations just above the extinction threshold (Hanski & Ovaskainen, 2002; Wallis-DeVries *et al.*, 2002). Currently, several sedentary host plant specialists are listed as critically endangered in Denmark, including the Marsh Fritillary (*Euphydryas aurinia*), Large Blue (*Maculinea arion*) and Lesser Marbled Fritillary (*Brenthis ino*) (Wind & Pihl, 2010). We therefore emphasize the urgent need to focus on host plant-specific management as a means to improve the conservation status of host plant constrained butterfly species (Tjørnløv *et al.*, early view). Such management efforts specifically tailored to improve host plant populations have already been recommended for endangered species such as the Marsh Fritillary (Smee *et al.*, 2011).

Trend for sedentary habitat specialists overwintering in the egg stage (cluster 2)

Sedentary habitat specialist species were often associated with forest habitats (Tables 2 and S6), including the Northern Chequered Skipper (*Carterocephalus silvicola*), Clouded Apollo (*Parnassius mnemosyne*) and Purple Emperor (*Apatura ilia*). Although several species within this group were associated with semi-natural grassland, including some species of high conservation concern, for example silver-spotted skipper (*Hesperia comma*), population trajectories for this group are likely to have been strongly influenced by historical forest management practices. During the 20th century, deciduous woodlands of lowland temperate Europe have become increasingly dense and shady due to the replacement of millennia-old management techniques by even-aged tall forest stands, with deleterious impacts for butterflies (Warren & Key, 1991; Bergman & Kindvall, 2004; Spitzer *et al.*, 2008). Furthermore, species-rich ecotones marking the transition zone between closed and open vegetation, and which often harbour high species diversity, including many forest butterflies, have increasingly been replaced by sharp borders. This has had detrimental effects on butterflies such as Ilex Hairstreak (*Satyrrium ilicis*) (Maes *et al.*, 2014). In Denmark, the emergence of intensified forestry occurred already in the late 1800s (Levin & Normander, 2008). This was reflected by the richness trajectory for habitat specialists, where species

richness declines at the 10 km × 10 km resolution were detected already in the 1920s and 30s (Fig. 1c, grey symbols). This was accompanied by increasing local homogenization, indicating the local extirpation of rare and spatially restricted species and range expansions of common and widespread species.

Because sedentary habitat specialists were mostly restricted to forest, and therefore occupied fewer grid cells than sedentary host plant specialists, local-scale species losses were generally less severe. However, regional-scale losses were more likely to occur and resulted in the net regional extinction of 27% of all species in this group (eight in total), including species such as Duke of Burgundy (*Hemearis lucina*, †1960), Scarce Heath (*Coenonympha hero*, †1982) and Ilex Hairstreak (†1997). Noticeably, all species which went extinct within this functional group were forest specialists, reflecting the degradation of forest habitats in Denmark. Like host plant specialists, low mobility is likely to have played an important role in the groups' decline, but because forest habitat degraded faster than semi-natural habitats, habitat specialist species were the first to be pushed towards regional extinction.

Trend for mobile generalists with mature overwintering stages (cluster 3)

Species of the third cluster were highly mobile (i.e. migratory), allowing species to permeate fragmented landscapes in their search for suitable habitat. A typical example species is the Green-veined White (*Pieris napi*), which relies on several widespread host plants, for example Garlic Mustard (*Alliaria petiolata*) and Cuckoo Flower (*Cardamine pratensis*), which are abundant even in intensively managed agricultural landscapes. Mobile generalist species have mature overwintering stages (i.e. adult or pupa), a trait which has been suggested to allow species to respond more readily to warmer spring temperatures or other environmentally favourable conditions (Diamond *et al.*, 2011; Breed *et al.*, 2013). In contrast, species which overwinter as eggs or larvae have been shown to be overrepresented among declining species, which may be linked to higher susceptibility to desiccation as seasons become warmer and drier (Breed *et al.*, 2013). Thus, overwintering as adult or pupa could constitute an important advantage for butterflies in the face of global change. However, both host plant- and habitat specialists lost diversity in the earliest time periods whereas mobile generalists did not start decreasing until more recently. This could imply that the latter group was affected by changing climates. Overall, our results highlight the need for more detailed studies of the importance of overwintering stage for butterfly species' resilience to climate change.

While the advantages associated with being a mobile generalist likely contributed to the relatively stable richness trends of cluster 3, declines were still detected. They were likely driven by factors such as decreasing availability of nectar resources or increased use of pesticides associated with agricultural intensification (European Environmental

Agency, 2013). In the light of the relatively small changes in species richness found for generalist assemblages, the signals of increasing biotic homogenization detected even in this group are noteworthy, suggesting a trend of increasing biotic homogenization due to the increasing predominance of broadly adapted and highly mobile butterfly species.

Outlook

With the recent rise in public participation in biological recording (i.e. citizen science), combined with the increasing effort to digitize historical species records, there has been a rapid increase in the availability of species occurrence data. Here, we compiled such a large, unstandardized dataset and used it to analyze trends in species richness and assemblage composition for Danish butterflies across multiple spatial and temporal scales. Our results draw attention to the possibility of documenting and understanding historical population trajectories not only of butterflies, but also of any other taxa for which historical occurrence data are available. We further emphasize that future research should compare long-term trends of Danish butterflies with other (e.g. EU) countries to better understand how different changes in land use and climate have affected the functional composition of butterfly communities across large spatial scales.

ACKNOWLEDGEMENTS

We sincerely thank Michael Stoltze, Gitte Calov, Martin Bjerg, Fugle og Natur, Birdlife Denmark, the Danish Lepidopterological Society and Entomologisk Selskab for Fyn for the use of their butterfly records. We also wish to thank Peder Klith Bøcher for guidance in GIS analyses. AE and TTH acknowledge funding from 15. Juni Fonden, WDK an University of Amsterdam (UvA) starting grant, LGC, OS and JCB support from the EU FP7 project STEP (grant no. 244090) and LGC and OS the EU FP7 project SCALES (Grant no. 226852).

REFERENCES

- Bartomeus, I., Ascher, J.S., Gibbs, J., Danforth, B.N., Wagner, D.L., Hedtkke, S.M. & Winfree, R. (2013) Historical changes in northeastern US bee pollinators related to shared ecological traits. *Proceedings of the National Academy of Sciences USA*, **110**, 4656–4660.
- Baselga, A. (2010) Partitioning the turnover and nestedness components of beta diversity. *Global Ecology and Biogeography*, **19**, 134–143.
- Bergman, K.O. & Kindvall, O. (2004) Population viability analysis of the butterfly *Lopinga achine* in a changing landscape in Sweden. *Ecography*, **27**, 49–58.
- Biesmeijer, J.C., Roberts, S.P.M., Reemer, M., Ohlemüller, R., Edwards, M., Peeters, T., Schaffers, A.P., Potts, S.G., Kleukers, R., Thomas, C.D., Settele, J. & Kunin, W.E. (2006) Parallel declines in pollinators and insect-pollinated plants in Britain and the Netherlands. *Science*, **313**, 351–354.
- Börschig, C., Klein, A.-M., von Wehrden, H. & Krauss, J. (2013) Traits of butterfly communities change from specialist to generalist characteristics with increasing land-use intensity. *Basic and Applied Ecology*, **14**, 547–554.
- Breed, G.A., Stichter, S. & Crone, E.E. (2013) Climate-driven changes in northeastern US butterfly communities. *Nature Climate Change*, **3**, 142–145.
- Bulman, C.R., Wilson, R.J., Holt, A.R., Bravo, L.G., Early, R.I., Warren, M.S. & Thomas, C.D. (2007) Minimum viable metapopulation size, extinction debt, and the conservation of a declining species. *Ecological Applications*, **17**, 1460–1473.
- Calinski, T. & Harabasz, J. (1974) A dendrite method for cluster analysis. *Communications in Statistics: Theory and Methods*, **3**, 1–27.
- Carvalho, L.G., Kunin, W.E., Keil, P., Aguirre-Gutiérrez, J., Ellis, W.N., Fox, R., Groom, Q., Hennekens, S., Van Landuyt, W., Maes, D., Van de Meutter, F., Michez, D., Rasmont, P., Ode, B., Potts, S.G., Reemer, M., Roberts, S.P.M., Schaminée, J., WallisDeVries, M.F. & Biesmeijer, J.C. (2013) Species richness declines and biotic homogenisation have slowed down for NW-European pollinators and plants. *Ecology Letters*, **16**, 870–878.
- Cattell, R.B. (1966) The Scree test for the number of factors. *Multivariate Behavioral Research*, **1**, 245–276.
- Chao, A., Chazdon, R.L., Colwell, R.K. & Shen, T.-J. (2005) A new statistical approach for assessing similarity of species composition with incidence and abundance data. *Ecology Letters*, **8**, 148–159.
- Colwell, R.K., Chao, A., Gotelli, N.J., Lin, S.-Y., Mao, C.X., Chazdon, R.L. & Longino, J.T. (2012) Models and estimators linking individual-based and sample-based rarefaction, extrapolation and comparison of assemblages. *Journal of Plant Ecology*, **5**, 3–21.
- Dapporto, L. & Dennis, R.L.H. (2013) The generalist–specialist continuum: testing predictions for distribution and trends in British butterflies. *Biological Conservation*, **157**, 229–236.
- Dennis, R.L. (2012) *A resource-based habitat view for conservation: butterflies in the British landscape*. John Wiley & Sons.
- Devictor, V., van Swaay, C., Brereton, T. *et al.* (2012) Differences in the climatic debts of birds and butterflies at a continental scale. *Nature Climate Change*, **2**, 121–124.
- Diamond, S.E., Frame, A.M., Martin, R.A. & Buckley, L.B. (2011) Species' traits predict phenological responses to climate change in butterflies. *Ecology*, **92**, 1005–1012.
- Dullinger, S., Essl, F., Rabitsch, W., Erb, K.-H., Gingrich, S., Haberl, H., Hülber, K., Jarošík, V., Krausmann, F., Kühn, I., Pergl, J., Pyšek, P. & Hulme, P.E. (2013) Europe's other debt crisis caused by the long legacy of future extinctions. *Proceedings of the National Academy of Sciences USA*, **110**, 7342–7347.

- European Environmental Agency (2013) *The European grassland butterfly indicator: 1990–2011*. Publications Office of the European Union, Luxembourg, pp. 34.
- Frederiksen, S.E., Rasmussen, F.N. & Seberg, O. (2006) *Dansk flora*. Gyldendal, Nordisk Forlag A/S, Copenhagen.
- García-Barros, E. (2000) Egg size in butterflies (Lepidoptera: Papilionoidea and Hesperioidea): a summary of data. *The Journal of Research on the Lepidoptera*, **35**, 90–136.
- Gotelli, N.J. & Colwell, R.K. (2001) Quantifying biodiversity: procedures and pitfalls in the measurement and comparison of species richness. *Ecology Letters*, **4**, 379–391.
- Hanski, I. & Ovaskainen, O. (2002) Extinction debt at extinction threshold. *Conservation Biology*, **16**, 666–673.
- Henle, K., Davies, K., Kleyer, M., Margules, C. & Settele, J. (2004) Predictors of species sensitivity to fragmentation. *Biodiversity & Conservation*, **13**, 207–251.
- Hill, M. & Smith, A. (1976) Principal component analysis of taxonomic data with multi-state discrete characters. *Taxon*, **25**, 249–255.
- Keil, P., Biesmeijer, J.C., Barendregt, A., Reemer, M. & Kunin, W.E. (2011) Biodiversity change is scale-dependent: an example from Dutch and UK hoverflies (Diptera, Syrphidae). *Ecography*, **34**, 392–401.
- Krauss, J., Bommarco, R., Guardiola, M., Heikkinen, R.K., Helm, A., Kuussaari, M., Lindborg, R., Öckinger, E., Partel, M., Pino, J., Poyry, J., Raatikainen, K.M., Sang, A., Stefanescu, C., Teder, T., Zobel, M. & Steffan-Dewenter, I. (2010) Habitat fragmentation causes immediate and time-delayed biodiversity loss at different trophic levels. *Ecology Letters*, **13**, 597–605.
- Kuussaari, M., Heliölä, J., Pöyry, J. & Saarinen, K. (2007) Contrasting trends of butterfly species preferring semi-natural grasslands, field margins and forest edges in northern Europe. *Journal of Insect Conservation*, **11**, 351–366.
- Lennon, J.J., Koleff, P., Greenwood, J. & Gaston, K.J. (2001) The geographical structure of British bird distributions: diversity, spatial turnover and scale. *Journal of Animal Ecology*, **70**, 966–979.
- Levin, G. & Normander, B. (2008) Arealanvendelse i Danmark siden slutningen af 1800-tallet. *Faglig rapport fra DMU nr. 682*, pp. 46. Danmarks Miljøundersøgelser, Aarhus Universitet.
- Loxdale, H.D., Lushai, G. & Harvey, J.A. (2011) The evolutionary improbability of ‘generalism’ in nature, with special reference to insects. *Biological Journal of the Linnean Society*, **103**, 1–18.
- Maes, D., Jacobs, I., Segers, N., Vanreusel, W., Van Daele, T., Laurijssens, G. & Van Dyck, H. (2014) A resource-based conservation approach for an endangered ecotone species: the Ilex Hairstreak (*Satyrium ilicis*) in Flanders (north Belgium). *Journal of Insect Conservation*, **18**, 939–950.
- McKinney, M.L. & Lockwood, J.L. (1999) Biotic homogenization: a few winners replacing many losers in the next mass extinction. *Trends in Ecology & Evolution*, **14**, 450–453.
- Menéndez, R., Megías, A.G., Hill, J.K., Braschler, B., Willis, S.G., Collingham, Y., Fox, R., Roy, D.B. & Thomas, C.D. (2006) Species richness changes lag behind climate change. *Proceedings of the Royal Society B: Biological Sciences*, **273**, 1465–1470.
- Miller-Rushing, A.J., Primack, R.B., Primack, D. & Mukunda, S. (2006) Photographs and herbarium specimens as tools to document phenological changes in response to global warming. *American Journal of Botany*, **93**, 1667–1674.
- Nekola, J.C. & White, P.S. (1999) The distance decay of similarity in biogeography and ecology. *Journal of Biogeography*, **26**, 867–878.
- Öckinger, E., Schweiger, O., Crist, T.O., Debinski, D.M., Krauss, J., Kuussaari, M., Petersen, J.D., Pöyry, J., Settele, J., Summerville, K.S. & Bommarco, R. (2010) Life-history traits predict species responses to habitat area and isolation: a cross-continental synthesis. *Ecology Letters*, **13**, 969–979.
- Olden, J.D., LeRoy Poff, N., Douglas, M.R., Douglas, M.E. & Fausch, K.D. (2004) Ecological and evolutionary consequences of biotic homogenization. *Trends in Ecology & Evolution*, **19**, 18–24.
- Parmesan, C. (2006) Ecological and evolutionary responses to recent climate change. *Annual Review of Ecology, Evolution, and Systematics*, **37**, 637–669.
- Pereira, H.M., Leadley, P.W., Proença, V. *et al.* (2010) Scenarios for global biodiversity in the 21st century. *Science*, **330**, 1496–1501.
- Potts, S.G., Biesmeijer, J.C., Kremen, C., Neumann, P., Schweiger, O. & Kunin, W.E. (2010) Global pollinator declines: trends, impacts and drivers. *Trends in Ecology & Evolution*, **25**, 345–353.
- Pöyry, J., Luoto, M., Heikkinen, R.K., Kuussaari, M. & Saarinen, K. (2009) Species traits explain recent range shifts of Finnish butterflies. *Global Change Biology*, **15**, 732–743.
- R Development Core Team (2013) *R: a language and environment for statistical computing*. R Foundation for Statistical Computing. Retrieved from <http://www.R-project.org>.
- Sang, A., Teder, T., Helm, A. & Partel, M. (2010) Indirect evidence for an extinction debt of grassland butterflies half century after habitat loss. *Biological Conservation*, **143**, 1405–1413.
- Schröter, D., Cramer, W., Leemans, R. *et al.* (2005) Ecosystem service supply and vulnerability to global change in Europe. *Science*, **310**, 1333–1337.
- Schweiger, O., Musche, M., Bailey, D., Billeter, R., Diekötter, T., Hendrickx, F., Herzog, F., Liira, J., Maelfait, J.-P., Speelmans, M. & Dziock, F. (2007) Functional richness of local hoverfly communities (Diptera, Syrphidae) in response to land use across temperate Europe. *Oikos*, **116**, 461–472.
- Smee, M., Smyth, W., Tunmore, M., French-Constant, R. & Hodgson, D. (2011) Butterflies on the brink: habitat requirements for declining populations of the marsh fritillary (*Euphydryas aurinia*) in SW England. *Journal of Insect Conservation*, **15**, 153–163.
- Spitzer, L., Konvicka, M., Benes, J., Tropek, R., Tuf, I.H. & Tufova, J. (2008) Does closure of traditionally managed open woodlands threaten epigeic invertebrates? Effects of coppicing and high deer densities. *Biological Conservation*, **141**, 827–837.

- Stefanescu, C., Carnicer, J. & Peñuelas, J. (2011) Determinants of species richness in generalist and specialist Mediterranean butterflies: the negative synergistic forces of climate and habitat change. *Ecography*, **34**, 353–363.
- Stoltze, M. (1996) *Danske dagsommerfugle*, Gyldendalske boghandel, Nordisk Forlag A/S, Copenhagen.
- Thomas, J.A. (2005) Monitoring change in the abundance and distribution of insects using butterflies and other indicator groups. *Philosophical Transactions of the Royal Society B: Biological Sciences*, **360**, 339–357.
- Tjørnløv, R., Kissling, W.D., Barnagaud, J.-Y., Bøcher, P. & Høye, T. (2015) Oviposition site selection of an endangered butterfly at local spatial scales. *Journal of Insect Conservation*, **19**, 377–391.
- Van Dyck, H., Van Strien, A.J., Maes, D. & Van Swaay, C.A.M. (2009) Declines in common, widespread butterflies in a landscape under intense human use. *Conservation Biology*, **23**, 957–965.
- WallisDeVries, M.F., Poschlod, P. & Willems, J.H. (2002) Challenges for the conservation of calcareous grasslands in northwestern Europe: integrating the requirements of flora and fauna. *Biological Conservation*, **104**, 265–273.
- Warren, M.S. & Key, R.S. (1991) Woodlands: past, present and potential for insects. *The Conservation of Insects and Their Habitats* (ed. by N.M. Collins), 155–212.
- Wilson, R.J., Thomas, C.D., Fox, R., Roy, D.B. & Kunin, W.E. (2004) Spatial patterns in species distributions reveal biodiversity change. *Nature*, **432**, 393–396.
- Wind, P. & Pihl, S. (2010) Den danske rødliste.-Danmarks Miljøundersøgelser, Aarhus Universitet, [2004].

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Figure S1 Multivariate analysis of species–trait relationships.

Table S1 Overview of data sources used for the analyses.

Table S2 Fifteen vagrant butterfly species not included in the analysis.

Table S3 Number of records from each time slice, which were included in the analysis.

Table S4 Number of cells included in species richness analysis at 10 km × 10 km (i.e. local) resolution.

Table S5 Comparison of species richness change between time periods, based on overlap of 95% confidence intervals.

Table S6 Overview of all 79 butterfly species and their life history traits used in the analysis.

Table S7 Statistical details of the effect of distance between cells and time period on species assemblage similarity analyses.

BIOSKETCH

Anne Eskildsen is a PhD student at Aarhus University's Department of Bioscience under the supervision of Toke T. Høye, Signe Normand and W. Daniel Kissling. Her main area of research is the long-term effects of land use and climate change on the distribution, richness and life history of butterflies. The team of co-authors is interested in macroecology, biogeography, species distributions, population ecology and global change.

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Editor: Ralph Mac Nally